

Incidence of treatment intensification based on HbA1c trends in people with type 2 diabetes

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ABSTRACT

Aims: Clinical inertia, the failure to intensify treatment despite unmet glycemic goals, is a key factor in poor glycemic management for type 2 diabetes. No studies have defined clinical inertia based on HbA1c trends. In this study, we aimed to assess treatment intensification according to HbA1c trends.

Materials and methods: We analyzed data from the Japan Diabetes Comprehensive Database Project based on an Advanced Electronic Medical Record System (J-DREAMS) between 2016 and 2023. Eligible patients with type 2 diabetes had (1) unchanged prescriptions for the past 90 days; (2) two consecutive consultations within 90 days; and (3) HbA1c levels $\geq 7\%$, or $\geq 7.5\%$, for patients aged ≥ 65 years at both consultations. Treatment intensification was defined as an addition or increased dose or switch in antidiabetic medications, inclusive of insulin, at the second consultation. Moreover, factors associated with treatment intensification were assessed.

Results: Of the 5,683 patients, 1,130 (19.9%) received intensified treatment at the second consultation. Intensification occurred more frequently with higher HbA1c levels or worsening HbA1c trends. However, treatments were not intensified in approximately 50% of the patients, with HbA1c levels $>8\%$ or a worsening of $>1\%$. Predictive factors included the HbA1c levels at the second consultation, changes in the HbA1c levels between the first and second consultations, the number of oral hypoglycemic medications, and the use of sulfonylureas or glinides.

Conclusions: Physicians should consider HbA1c trends to guide treatment intensification when HbA1c levels exceed target thresholds. Clinical inertia remains an important issue in diabetes management.

INTRODUCTION

Diabetes affects approximately 537 million adults worldwide (10.5% of those aged 20–79 years), with type 2 diabetes comprising $>90\%$ of cases¹. Despite guideline targets of HbA1c $<7\%$ to reduce micro- and macrovascular risk, many patients with type 2 diabetes remain above goal^{2–6}. This gap reflects disease progression and limited pathophysiology-directed options, alongside patient, treatment, and system constraints^{7–13}.

Clinical inertia is defined as the failure to intensify treatment despite not achieving glycemic targets¹³. It has been identified as a principal physician-related factor contributing to poor glycemic management. To prevent clinical inertia, the American Diabetes Association (ADA) and Japan Diabetes Society (JDS) recommend reassessing and modifying diabetes treatment every 3–6 months^{14,15}. Nonetheless, treatment intensification is frequently delayed^{16–19}.

Pantalone *et al.* reported that more than 60% of patients receiving two oral antidiabetic medications did not undergo

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treatment intensification within 6 months, particularly when the HbA1c level exceeded 7%¹⁶. Similarly, over a 4-year observation period, Mata-Cases *et al.* found that 20% of patients treated with two or more oral antidiabetic medications, despite exhibiting an HbA1c level of >8%, did not undergo treatment intensification. Furthermore, the median time to first treatment intensification was reported to be 17.1 months¹⁷. Khunti *et al.* demonstrated that the initiation of additional therapy—bolus insulin, premixed insulin, or a glucagon-like peptide-1 receptor agonist (GLP-1RA)—in patients receiving basal insulin was delayed for >4 years¹⁸. In Japan, during a 1.5-year observation period, only 21% of patients with type 2 diabetes receiving basal insulin were transitioned to an intensified insulin regimen¹⁹.

In these studies, single-point HbA1c measurements have been utilized to evaluate the effectiveness of glycemic management, with clinical inertia defined as the failure to intensify treatment despite not achieving the target HbA1c level. In particular, when the target HbA1c level is not met, further intensification may be deemed unnecessary based on previous trends. For example, when an HbA1c target level of <7% is applied, a patient whose HbA1c level decreases from 8.1% to 7.1% within 3 months without any modification to treatment would still be classified as exhibiting clinical inertia. However, as HbA1c levels frequently decrease with a delay in treatment intensification, physicians most likely view treatment intensification as unnecessary. Thus, defining clinical inertia based on HbA1c trends may allow for more accurate identification of true clinical inertia and high-risk patient populations. To the best of our knowledge, no studies have defined clinical inertia according to HbA1c trends.

Therefore, in this study, data from the JDS-certified specialized diabetic facilities were utilized to examine the incidence of treatment intensification based on HbA1c trends. Furthermore, the factors associated with treatment intensification were investigated. Based on these findings, the population exhibiting clinical inertia—defined as the failure to intensify treatment despite inadequate glycemic control—was more precisely characterized.

MATERIALS AND METHODS

Ethical approval and informed consent

This study was approved by the Ethics Committee (EC) of the National Center for Global Health and Medicine (currently the Japan Institute for Health Security), Tokyo, Japan (NCGM-S-001702-22). It was conducted in accordance with the principles of the Declaration of Helsinki. An opt-out procedure was carried out, as instructed by the institutional review board, to permit the use of electronic medical record data.

Data curation

In this retrospective cohort study, data were obtained from the Japan Diabetes Comprehensive Database Project based on an Advanced Electronic Medical Record System (J-DREAMS). J-DREAMS provides access to the medical

records of patients with type 2 diabetes attending hospitals designated as educational facilities by the JDS²⁰. Type 2 diabetes was diagnosed according to the criteria established by the JDS²¹. Therefore, patients with type 2 diabetes who were registered in this database between January 1, 2016, and November 15, 2023, were included in the study. The database contains information regarding the type of diabetes, age at diagnosis, comorbidities, smoking and drinking history, and microvascular complications. Blood test results and prescription data were also extracted from the electronic medical records.

Study design

Based on previous literature^{16,22}, patients with type 2 diabetes requiring treatment intensification were defined as follows: (1) those whose prescriptions had not changed in the past 90 days; (2) those who had two consecutive outpatient consultations within 90 days; and (3) those with a HbA1c level of $\geq 7\%$, or $\geq 7.5\%$ for patients aged ≥ 65 years, at both consultations (Figures 1 and S1). The observation period commenced on the earliest date between January 1, 2016, and November 15, 2023, when both the HbA1c levels and prescription patterns simultaneously fulfilled the inclusion criteria.

Meanwhile, patients (1) aged ≤ 18 or ≥ 80 years; (2) with an HbA1c level of <7%, or <7.5% for those aged ≥ 65 years; (3) who had been prescribed an oral hypoglycemic medication from another hospital; and (4) who had been prescribed insulin or GLP-1RA injection products were excluded. The reason why patients who had been prescribed insulin or GLP-1RA were excluded was to focus on intensifying treatment among people who were only taking oral medication, because it seemed to be more difficult to add another medication to a patient who is taking only oral medication than to change the dose for a patient who is already using an injectable medication. Moreover, it was impossible to determine from the J-DREAMS data whether the dose of insulin or GLP-1RA injection products were changed.

Treatment intensification

Based on the HbA1c level obtained from the blood test result and treatment prescribed at the second consultation, treatment intensification was defined as the (1) addition, (2) dose escalation, (3) or switching of antidiabetic medications, including insulin. Treatment intensification was not considered if the medication remained unchanged or the dose was reduced (Figure S1).

Predictive factors and covariates

In this study, the main predictive factors included changes in the HbA1c levels between the first and second consultations, in addition to the HbA1c level recorded at the second consultation. HbA1c levels were measured 0–28 days before the first and second consultations and again within 28–98 days following the third consultation.

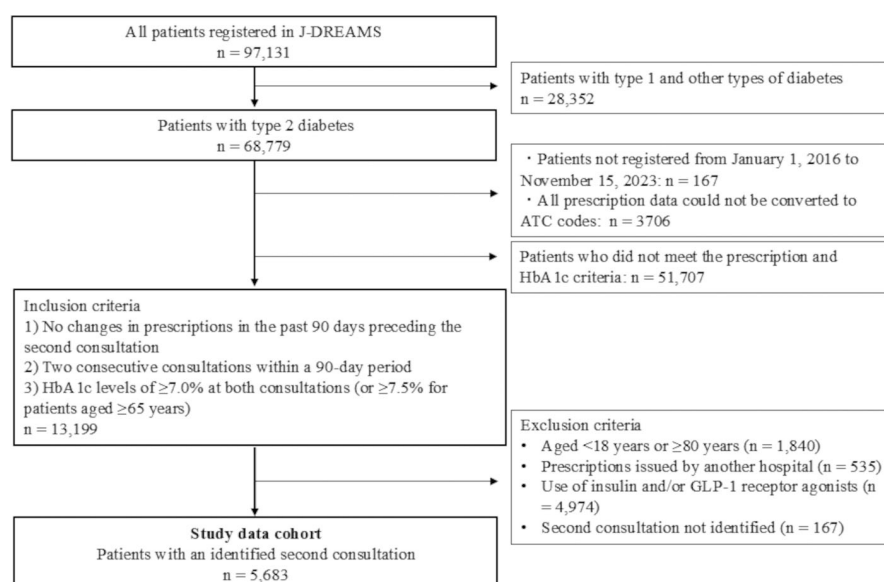


Figure 1 | Flowchart illustrating the patient selection process. ATC codes, Anatomical Therapeutic Chemical classification codes; GLP-1 receptor agonist, glucagon-like peptide-1 receptor agonist; J-DREAMS, Japan Diabetes Comprehensive Database Project Based on an Advanced Electronic Medical Record System.

The covariates included age, sex, height, body weight, smoking status, drinking history, microvascular complications, laboratory data, and treatment information. These data were systematically collected during the first consultation. Laboratory parameters, including the high-density lipoprotein and low-density lipoprotein cholesterol, triglyceride, and blood glucose levels, were measured 0–28 days prior to the first consultation and used as baseline data. Medication prescriptions were assumed to have been issued on or after the date of each consultation.

Statistical analysis

Patient selection and baseline characteristics, including age, duration since diabetes diagnosis, number of oral hypoglycemic agents, comorbidities, and HbA1c levels, were initially analyzed. Comparisons were conducted based on the presence or absence of treatment intensification. The *t*-test was used for assessing continuous variables, whereas the χ^2 or Fisher's exact tests were applied to analyze categorical variables. For the sensitivity analysis, data were stratified by age, with subgroup analyses performed separately for patients aged <65 years and ≥65 years. A subcohort analysis was also conducted to evaluate treatment intensification exclusively in patients prescribed sulfonylureas (SUs) or glinides.

Next, the association between the HbA1c level at the second consultation and the changes in HbA1c levels between the first and second consultations was examined. Logistic regression analysis was used to identify the factors associated with treatment intensification.

Finally, HbA1c trends were evaluated in 4,656 eligible patients for whom HbA1c measurements were available at the third consultation. A *P*-value of <0.05 was considered statistically significant. All statistical analyses were performed using STATA/SE version 16.1 (Stata Corp., College Station, TX, USA).

RESULTS

Patient baseline characteristics

This study included 5,683 patients with type 2 diabetes whose prescriptions remained unchanged for the past 90 days and had an HbA1c level of ≥7% or ≥7.5% (for those aged ≥65 years; Figures 1 and S1). The median HbA1c level at the first consultation was 7.6%, with a median interval of 56 days between the first and second consultations (Tables 1 and S1). Nearly all patients (98.5%) were treated with oral hypoglycemic agents, most commonly either as monotherapy or in combination with two to three additional medications. Diabetic retinopathy was present in 10% of patients, whereas 1% had an estimated glomerular filtration rate of <30 mL/min/1.73 m² or were receiving renal replacement therapy. A small proportion of patients presented with severe liver dysfunction. Approximately 60% of patients were prescribed medications for hypertension, dyslipidemia, or both.

Among the total cohort, 1,130 patients (19.9%) underwent treatment intensification at the second consultation and were included in the intensified treatment cohort whereas 4,553 patients (80.1%) did not receive intensified treatment and were classified as the non-intensified treatment cohort (Table 1). The

Table 1 | Demographic and clinical characteristics of the study participants

	Total study population <i>n</i> = 5,683 ^a	Intensified treatment cohort <i>n</i> = 1,130 (19.9%) ^a	Non-intensified treatment cohort <i>n</i> = 4,553 (80.1%) ^a	<i>P</i> ^b
Age, years	62 (54–71)	64 (54–71)	62 (54–71)	0.04
Female patients, <i>n</i> (%)	2085 (36.7)	398 (35.2)	1,687 (37.1)	0.25
Time since diabetes diagnosis, years	11 (6–17), <i>n</i> = 2,994	11 (6–17), <i>n</i> = 640	11 (6–17), <i>n</i> = 2,354	0.70
HbA1c level at the 1st consultation, %	7.6 (7.3–8.0)	7.7 (7.5–8.0)	7.6 (7.3–8.0)	0.15
HbA1c level at the 2nd consultation, %	7.7 (7.4–8.1)	7.9 (7.6–8.4)	7.6 (7.3–8.0)	<0.001
BMI, kg/m ²	26.6 ± 5.1, <i>n</i> = 4,755	26.6 ± 5.2, <i>n</i> = 944	26.6 ± 5.1, <i>n</i> = 3,811	0.91
Number of oral antidiabetic medications, <i>n</i> (%)				
0	83 (1.5)	14 (1.2)	69 (1.5)	0.001
1	1,226 (21.6)	287 (25.4)	939 (20.6)	
2	1,774 (31.2)	374 (33.1)	1,400 (30.8)	
3	1,628 (28.7)	283 (25.0)	1,345 (29.5)	
4	764 (13.4)	138 (12.2)	626 (13.8)	
5	178 (3.1)	28 (2.5)	150 (3.3)	
6	30 (0.5)	5 (0.5)	24 (0.5)	
Comorbidities				
Cardiovascular disease, <i>n</i> (%)				
Yes	413 (7.3)	78 (6.9)	335 (7.4)	0.47
No	2,710 (47.7)	568 (50.2)	2,142 (47.1)	
Unknown	39 (0.7)	10 (0.9)	29 (0.6)	
Missing	2,521 (44.4)	474 (42.0)	2,047 (45.0)	
Stroke, <i>n</i> (%)				
Yes	238 (4.2)	38 (3.4)	200 (4.4)	0.14
No	2,952 (51.9)	623 (55.1)	2,329 (51.2)	
Unknown	26 (0.5)	4 (0.4)	22 (0.5)	
Missing	2,467 (43.4)	465 (41.2)	2,002 (43.9)	
Peripheral artery disease, <i>n</i> (%)				
Yes	78 (1.4)	17 (1.5)	61 (1.3)	0.54
No	3,039 (53.5)	628 (55.6)	2,411 (53.0)	
Unknown	60 (1.1)	9 (0.8)	51 (1.1)	
Missing	2,506 (44.1)	476 (42.1)	2,030 (44.6)	
Lower limb amputation, <i>n</i> (%)				
Yes	11 (0.2)	3 (0.3)	8 (0.2)	0.81
No	3,172 (55.8)	654 (57.9)	2,518 (55.3)	
Unknown	21 (0.4)	5 (0.4)	16 (0.4)	
Missing	2,479 (43.6)	468 (41.4)	2,011 (44.2)	
Severe hypoglycemia, <i>n</i> (%)				
Yes	19 (0.3)	5 (0.4)	14 (0.3)	0.11
No	2,082 (36.6)	419 (37.1)	1,663 (36.5)	
Unknown	539 (9.5)	130 (11.5)	409 (9.0)	
Missing	3,043 (53.6)	576 (51.0)	2,467 (54.2)	

Abbreviation: BMI, Body Mass Index. ^aContinuous variables are expressed as the means ± SD or medians and interquartile ranges (25th and 75th percentiles). Categorical variables are expressed as *n* (%). ^bStatistical significance is set at a *P*-value of <0.05.

demographic characteristics were generally similar between patients who received treatment intensification and those who did not. However, the intensified treatment cohort was prescribed fewer oral hypoglycemic agents and exhibited higher HbA1c levels at the second consultation compared with the non-intensified treatment cohort.

The sensitivity analysis yielded broadly consistent results between patients aged <65 and ≥65 years. Notably, HbA1c levels at the first consultation were significantly higher in the intensified treatment cohort compared with the non-intensified treatment cohort (Table S2). A subgroup analysis was conducted in patients prescribed SUs or glinides (Table S3). Of the

2,300 patients receiving either medication, 18.4% underwent treatment intensification, whereas 81.6% did not, reflecting the overall intensification rates observed in Table 1.

Treatment intensification

HbA1c levels at the second consultation and the changes in HbA1c levels between the first and second consultations were assessed. Patients whose HbA1c levels remained elevated but had improved at the second consultation were less likely to receive intensified treatment. Conversely, those with persistently high or worsening HbA1c levels were more likely to receive intensified treatment (Figure 2a). Similar findings were observed when patients were stratified by age (Figure 2b,c). Of the patients with a second HbA1c level of $\geq 8\%$ or an increase of $>1\%$ from the first to the second consultation, approximately half did not receive intensified treatment.

Logistic regression analysis was performed to identify factors predicting treatment intensification. The HbA1c level at the second consultation and changes in the HbA1c levels between the first and second consultations were significant predictors. The odds ratio (OR) for the HbA1c level at the second consultation was 1.33 (95% CI: 1.21–1.46, $P < 0.001$); conversely, the OR for the change in HbA1c levels between the first and second consultations was 2.64 (95% CI: 2.25–3.12, $P < 0.001$; Table S4).

The likelihood of treatment intensification was found to decrease as the number of oral hypoglycemic medications increased. By contrast, treatment intensification was more likely in patients who were not prescribed SUs or glinides. When compared with those prescribed a single antidiabetic agent, the

OR for treatment intensification was 0.58 (95% CI: 0.56–0.80, $P < 0.001$) in patients receiving three or four antidiabetic medications and was 0.58 (95% CI: 0.39–0.87, $P < 0.001$) in those receiving five or six antidiabetic medications. Furthermore, patients not prescribed SU and glinide were more likely to receive intensified treatment, with an OR of 1.34 (95% CI: 1.15–1.55, $P < 0.001$).

HbA1c trends

Figure 3 presents the HbA1c trends among the 4,656 patients whose HbA1c levels were assessed at the third consultation. The median differences in HbA1c levels between the second and third consultations were -0.33 in the intensified treatment cohort and -0.04 in the non-intensified treatment cohort ($P < 0.001$). The median (interquartile range) HbA1c level at the third consultation was 7.6 (7.2–8.0) in both cohorts. These findings suggest that glycemic targets were not achieved in a substantial proportion of patients.

DISCUSSION

This study found that treatment intensification was frequently not implemented appropriately, particularly among patients whose HbA1c levels were not markedly elevated and demonstrated improvement. Patients prescribed more than three antidiabetic medications or those receiving SUs or glinides were less likely to receive intensified treatment. To our knowledge, this is the first study to evaluate treatment intensification in relation to preceding HbA1c trends. These findings allowed for the evaluation of clinical inertia in a real-world setting based on HbA1c trends.

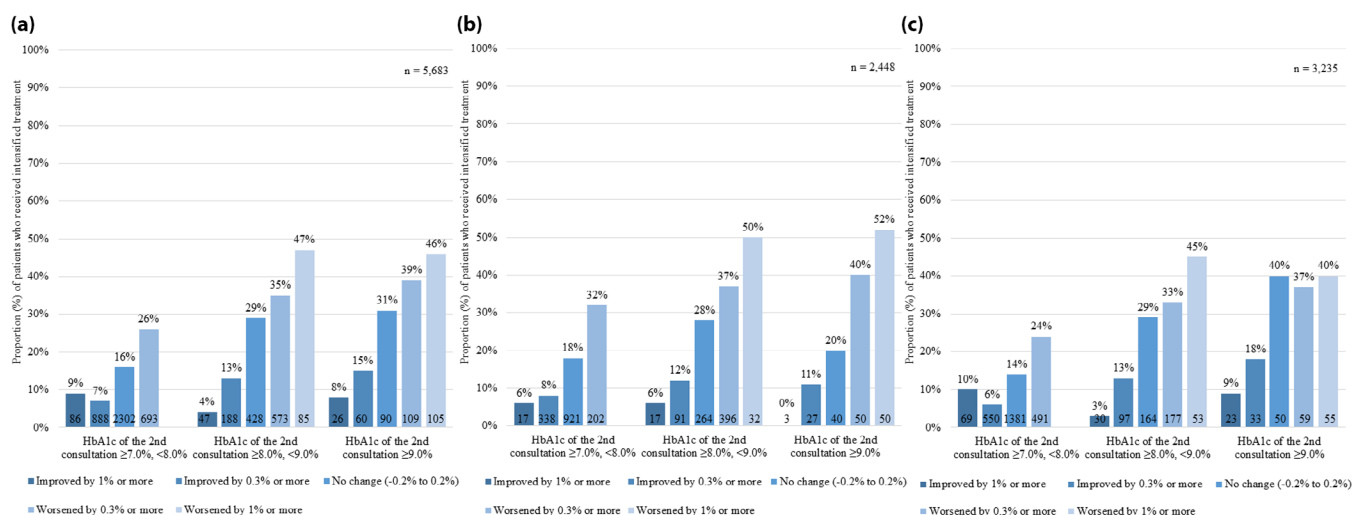


Figure 2 | Treatment intensification rates based on HbA1c levels recorded at the second consultation and changes in HbA1c levels between the first consultation and second consultation are presented: (a) All patients; (b) Patients aged ≥ 65 years; (c) Patients aged < 65 years. Numbers at the base of each bars indicate the total number of patients, whereas the percentages above the bars depict the proportion of patients who underwent treatment intensification.

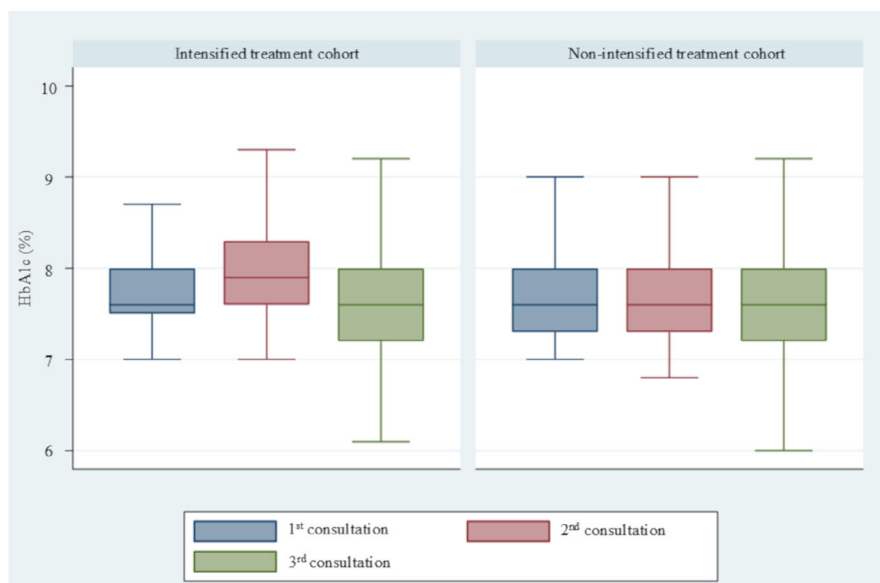


Figure 3 | Box plots of HbA1c levels determined by consultation and cohort.

Although numerous studies have investigated clinical inertia—defined as the failure of treatment intensification despite sub-optimal glycemic control—many have relied on a single HbA1c measurement for their assessment. However, the assessment of clinical inertia based on an HbA1c measurement may not accurately reflect the complexities of clinical decision-making in practice. Previous studies have reported the occurrence of clinical inertia in 26–72% of patients^{16,17,23,24}; however, the present study demonstrated a higher prevalence (80.1%) when HbA1c trends were considered.

Several factors explain the high rate of clinical inertia and the limited implementation of treatment intensification. As illustrated in Figure 2, treatment intensification was less frequently undertaken when HbA1c levels showed improvement, even if the levels remained elevated between the first and second consultations. Many physicians likely believed that patients with elevated but improving HbA1c levels would eventually achieve their glycemic targets without modifications to the medication regimen. In such cases, if only a single HbA1c value was considered, treatment intensification might be necessary. However, the need for treatment intensification cannot be accurately determined without assessing glycemic progress and the likelihood of achieving treatment goals.

Second, treatment intensification was assessed over a short period, whereas previous studies evaluated this process over longer durations. The ADA and JDS recommend reassessing treatment every 3–6 months; nonetheless, treatment intensification may, in practice, require a longer timeframe. Moreover, differences in the definition of treatment intensification across studies may have influenced the reported incidence. In the present study, treatment intensification was defined as the

addition, dose escalation, or switching of antidiabetic medications, including insulin. However, previous studies, such as that by Pantalone *et al.*, included dietary consultations with a nutritionist and medication titration as components of treatment intensification²⁵.

Clinical inertia may result from factors related to the patient, physician, and healthcare system^{26–28}. In this study, patients who had visited a hospital within the past 90 days and had not experienced any changes to their prescriptions were examined. As the J-DREAMS database comprises only specialized diabetes care facilities, physician-related factors such as the lack of knowledge of clinical guidelines or unfamiliarity with the latest treatment options were considered less likely contributors to inertia compared to general practice settings. Similarly, healthcare system-related factors, including difficulties in prescribing the latest treatments, were less likely to contribute to inertia. However, patient-related factors, such as the presence of multiple complications and polypharmacy, were more likely to be present. Likewise, physician-related factors, including insufficient consultation time and an overburdened work environment, were more likely to contribute to inertia.

Financial barriers are among the most significant patient-related obstacles. In a survey conducted in Japan, many diabetologists expressed concerns regarding the economic challenges associated with initiating insulin therapy²⁹. Financial barriers are also considered healthcare system-related factors, because they are associated with the reimbursement structure. In Japan, patients need to pay a certain amount for medication, and the more medicines they take, the greater the financial burden. In the present study, patients who did not receive intensified treatment were prescribed a greater number of oral

hypoglycemic agents than those who did. Concerns regarding the increased financial burden associated with initiating new medications or escalating doses may have hindered the implementation of treatment intensification. Moreover, when starting new medications such as injection therapy, instructions on how to use them are required. However, consultation time is limited. Many physicians might consider the explanations time-consuming²⁹. These are considered to be health-care system-related factors.

The intensified treatment cohort had higher HbA1c levels at the second consultation compared with the non-intensified treatment cohort. However, approximately half of the patients with elevated and worsening HbA1c levels did not receive intensified treatment (Figure 2). In some cases, non-pharmacologic strategies—such as dietary review, enhanced exercise regimens, or adjustments in the timing of administration—may have been employed. However, we could not determine from the J-DREAMS data whether lifestyle counseling and adherence interventions were provided. Then if we could consider these non-pharmacologic strategies, the percentage of clinical inertia was possibly lower like previous literature. Whereas HbA1c levels in the non-intensified treatment cohort remained unchanged, and the intensified treatment cohort showed a greater improvement in HbA1c levels at the third consultation compared with the second. These findings suggest that appropriate treatment intensification, such as pharmacologic adjustments and lifestyle interventions, was likely not implemented in the non-intensified treatment cohort.

The ADA advocates individualizing blood glucose targets based on patient age and hypoglycemia risk, particularly recommending the discontinuation of SUs and glinides due to the risk of hypoglycemia^{9,30}. In Japan, a more relaxed HbA1c target has been recommended, particularly for older adults using SUs or glinides^{31,32}. The risk of hypoglycemia may increase during treatment intensification^{26,27}, particularly among older adults, those with a long duration of diabetes, and those with multiple comorbidities. Both the ADA and JDS recommend individualized HbA1c goals tailored to each patient's clinical profile, with higher targets suggested for those at a higher risk of hypoglycemia⁹. Treatment intensification is less likely to be implemented as patient age increases³³. However, in the present study, patients in the intensified treatment cohort were slightly older than those in the non-intensified treatment cohort. Moreover, the treatment intensification rate was higher in patients aged ≥ 65 years than among those aged < 65 years (23.0% and 17.5%, respectively). This difference was likely influenced by the HbA1c inclusion criteria, which were defined as $\geq 7.0\%$ for patients aged < 65 years and $\geq 7.5\%$ for those aged ≥ 65 years.

In addition, several factors were identified as predisposing patients to treatment intensification, including a high HbA1c level at the second consultation, a rapidly rising HbA1c trajectory, and the absence of SU or glinide administration. Both elevated HbA1c levels at the second consultation and a rapidly increasing HbA1c level were strong predictors of treatment

intensification. This finding indicates that a single HbA1c measurement alone is insufficient for accurately assessing clinical inertia. However, treatment intensification was less frequently implemented in patients in their 50s. Previous studies have reported that glycemic control tends to be poorer among individuals aged 45–64 years than among those aged ≥ 65 years³⁴. Additionally, higher age is a predictive factor for clinical inertia among adults aged 18–65 years²⁴. Although the underlying cause remains unclear, lifestyle-related factors, such as occupational responsibilities and time constraints, may contribute to this trend.

This study has some limitations. We excluded patients already using GLP-1RA or insulin at baseline because it was impossible to determine whether the dose of insulin or GLP-1RA was changed. These medications are increasingly common in clinical practice; therefore, the impact on inertia may not be accurately assessed. As recommended by the ADA, treatment targets should be individualized based on patient comorbidities and prognosis⁹; however, such information was not available in the J-DREAMS database. Moreover, data on diet and exercise interventions were not systematically recorded. Many patients who required intensified treatment may have initially received recommendations to improve their diet and physical activity. Whether adherence to oral medication may have improved through adjustments in administration timing or by consolidating medications into single packages remains unclear. The small number of patients not receiving oral hypoglycemic agents limited the accurate assessment of how the number of medications influenced inertia. The logistic regression model did not include several covariates such as comorbidities, renal function, and socioeconomic factors, which may have caused confounding.

In conclusion, this is the first study to examine clinical inertia in practice while accounting for HbA1c trends and the associated factors. Clinical inertia was frequently observed in patients with high HbA1c levels at specialized facilities for type 2 diabetes in Japan. This included patients whose treatment was not intensified despite persistently high HbA1c levels. The determination of clinical inertia proved challenging, as it required a comprehensive evaluation of multiple clinical factors in each patient. Nonetheless, a substantial number of patients exhibited high and worsening HbA1c levels without corresponding treatment intensification. Clinical inertia remains an important issue that must be addressed by health-care providers. Establishing individualized HbA1c targets at each consultation and reassessing treatment based on trends are essential for achieving optimal glycemic control.

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DISCLOSURE

The conflicts of interest for Mitsuru Ohsugi are as follows: reports receiving honoraria for lectures from Astellas Pharma Inc., MSD K.K., Otsuka Pharmaceutical Co., Ltd., Kissei Pharmaceutical Co., Ltd., Teijin Pharma Ltd., Nippon Boehringer Ingelheim Co., Ltd., Sanofi K.K., Abbott Japan LLC, Sumitomo Dainippon Pharma Co., Ltd., Novo Nordisk Pharma Ltd., Kowa Co., Ltd., AstraZeneca K.K., Taisho Pharmaceutical Co., Ltd., Daiichi Sankyo Co., Ltd., Mitsubishi Tanabe Pharma Corporation, Ono Pharmaceutical Co., Ltd., Eli Lilly Japan K.K., Takeda Pharmaceutical Co., Ltd. Novartis Pharma K.K., and Bayer Yakuhii Ltd.; clinical commissioned/joint research grants from Novo Nordisk Pharma Ltd., Eli Lilly Japan K.K., Nippon Boehringer Ingelheim Co., Ltd., Sanofi K.K., Abbott Japan LLC, MSD K.K., Kyowa Kirin Co., Ltd., Astellas Pharma Inc., Sumitomo Dainippon Pharma Co., Ltd. and Bayer Yakuhii Ltd., and scholarship grants from Sumitomo Dainippon Pharma Co., Ltd. and Nippon Boehringer Ingelheim Co. Other authors declare no conflict of interest.

Approval of the research protocol: This study was approved by the Ethics Committee (EC) of the National Center for Global Health and Medicine (currently the Japan Institute for Health Security), Tokyo, Japan.

Informed Consent: An opt-out procedure was carried out, as instructed by the institutional review board, to permit the use of electronic medical record data.

Approval date of Registry and the Registration No. of the study: November 2024, NCGM-S-001702-22.

Animal Studies: N/A.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1. Graphical representation of longitudinal study.

Table S1. Characteristics of study participants.

Table S2. Summary of baseline participants' characteristics with and without treatment intensification by age.

Table S3. Participants' characteristics with and without treatment intensification only in cases receiving sulfonylurea or glinide.

Table S4. Logistic regression analysis of factors for treatment intensification.